

Raihan Rafeek

rafeekrn@mail.uc.edu | linkedin.com/in/raihan-rafeek | github.com/rai1975 | rai-1975.com

EDUCATION

University of Cincinnati, Cincinnati, OH

Bachelor of Science in Computer Science

GPA: 3.97

May 2027

EXPERIENCE

LeopardSAT - 1 ADCS Developer

Aug 2024 - Oct 2025

CubeCats - University of Cincinnati

Cincinnati, OH

- **Lead and directed** team building **simulation software** for CubeSat project in partnership with **NASA and Nanoracks**
- Applied physics models of magnetic hysteresis to improve detumbling efficiency and stability
- Optimized 6M+ point **dataset preprocessing pipeline, reducing simulation runtime by 30%**
- **Ohio's first successful student led cube-sat project** scheduled to launch into orbit April 2026

Undergraduate Research Assistant

Aug 2024 - Jun 2025

University of Cincinnati

Cincinnati, OH

- Assisted in the design of algorithms for **feature extraction from high-dimensional datasets** using TDA
- Evaluated the efficiency of **PCA, t-SNE, UMAP** to enhance downstream model performance
- Implemented testing pipeline to benchmark the time complexity of the algorithm
- **Co-authored paper** on Topological Study of β -Sparsified d-Uniform Hypergraph Based Simplicial Complexes

AI & Data Science Intern

Jan 2025 - Apr 2025

Exelixa AI

Dubai, UAE

- Deployed computer-vision attendance system, cutting manual **1-hour process to real-time with 98% efficiency gain**
- Developed AI-driven automation pipelines and interactive workflows for 5 client projects
- Integrated ML prediction models (PyTorch, scikit-learn, LangChain) to meet custom requirements
- Created high-impact demos showcased at AI Everything 2025 and other conferences in the UAE

Software Engineering Intern

Fall 2023, Summer 2024

Idaho National Laboratory

Idaho Falls, ID

- Contributed to the development of **safety-critical software for nuclear inventory management** using **Ruby on Rails and PostgreSQL**, ensuring compliance with strict regulatory standards and minimizing risk
- Implemented major changes to improve search result filtering speed by optimizing queries through a **database migration and implementing caching**
- **Resolved 40+ bugs, including 3 critical fixes**, new features, and UI/UX inconsistencies ensuring smooth and reliable application functionality

PROJECTS

Gyrus | *Python, LangChain, CrewAI, Exa, Electron.js, Neo4j*

July 2025

- Developed a browser that uses AI to promote thinking as opposed to cognitive offloading
- Implemented novel **user-intent-detection model ensemble** using Roberta and LangChain, achieving **accuracy of 0.94**
- Designed and developed Neo4j graph database to store user queries, map previous intentions and history
- Implemented Exa and CrewAI agents to handle web crawling and search based on intent

RentRadar | *Python, Flask, LangChain, React, MaterialUI, PostgreSQL*

May 2025 - Aug 2025

- **Designed and built a full-stack web application**, managing both frontend and backend components
- Deployed and maintained the application in a production environment, ensuring stability and performance
- **Supported 50+ active users**, continuously improving functionality based on feedback and usage data
- Won 2nd Place at the 2025 CEAS Summer Innovation Challenge, raising over \$2,000

WhiteBox | *Python, Neo4j, Llama 3.1, HuggingFace, Docker, ReactJS*

Sep 2024 - Oct 2024

- Developed architecture for RAG using GraphDB to store and navigate concept maps of text data
- **Designed database to handle over 70,000 nodes with 140,000 edges** for conceptual queries beyond raw vector searches
- Created a pipeline leveraging **fine-tuned large language models** to visualize relationships between entities
- Implemented advanced querying techniques for efficient data retrieval in the GraphDB framework
- Won Best Hack at the Midwestcon - Future of Data Hackathon with over 70 teams

TECHNICAL SKILLS

Languages: Python, Ruby, JavaScript, TypeScript, C, C++, C#, Java

Frameworks & Libraries: Flask, Rails, Angular, ReactJS, Electron.js, Pandas, NumPy, Matplotlib, Scikit-learn, SciPy

Databases & Query Languages: MySQL, Postgres, Cypher

AI & Machine Learning: BERT, LLaMA, OpenAI, Torch, TensorFlow, HuggingFace Transformers, LangChain

Tools & Scripting: Git, Docker, Postman, Zsh, Bash, PowerShell

Creative & Hardware Skills: Adobe Illustrator, Digital Audio Workstation (DAW), Woodworking, 3-D Printing, Laser Cutting

ACHIEVEMENTS

Scholarships: NEXT Innovation Scholar, CEAS International Outreach Scholar

Awards: Dean's List, 3x Hackathon Winner, Raised \$500 at UC Startup Weekend 2024